Perishable Lot–Sizing with Heterogeneous Shelf Lives

Compact MIP with Last–Expiry–First Allocation

Indices and sets

- $T = \{1, 2, \ldots, \mathbf{T}\}$ — planning periods
- $I = \{1, 2, \ldots, \mathbf{I}\}$ — set of items
- $t \in T$ — production period, $u \in T$ — consumption period
- m_{it} — shelf life of item i produced in period t (may vary by t)

For each item i and consumption period u, let

$$P_{iu} := \{\, t \in T : t \leq u \leq t + m_{it}\}$$

be the set of feasible production periods. For each $t \in P_{iu}$ define the expiry date: $e_{itu} = t + m_{it}$. Denote by,

$$e_{iu}^{(0)} \geq e_{iu}^{(1)} \geq \cdots \geq e_{iu}^{(G_{iu}-1)}$$

the distinct values of e_{itu}, sorted from largest (latest expiry) to smallest (earliest expiry), and write G_{iu} for the number of distinct expiry dates. For each index g with $0 \leq g < G_{iu}$, define:

$$\mathcal{G}_{iug} = \{\, t \in P_{iu} : e_{itu} = e_{iu}^{(g)}\}.$$

Thus $\mathcal{G}_{iu0}$ contains those production periods whose lots expire last, whereas $\mathcal{G}_{iuG_{iu}-1}$ contains those that expire first.

Perishability

A lot produced at period t may be used only in periods $u \geq t$ satisfying $u - t \leq m_{it}$. Such a lot expires at period $t + m_{it}$.

Parameters

d_{iu}	demand of item i in period u
c_{it}	unit production cost of item i in period t
h_{it}	holding cost per unit of age (constant or time-varying)
s_{it}	setup cost for item i in period t
κ_t	total production capacity available in period t
p_{it}	optional per-item production capacity in period t
μ_{it}	big-M constant for setup linking, defined as $\mu_{it} := \sum_{u \in P_{iu}} d_{iu}$
g_{itu}	cost of producing in t and consuming in u, namely $c_{it} + h_{it}\,(u - t)$
C_{iu}	upper bound on the quantity allocated to demand (i, u) (in a no-backorder model, $C_{iu} = d_{iu}$)
ε	small positive constant used in the sufficiency cuts

Decision variables

$X_{itu} \geq 0$	quantity of item i produced in period t and consumed in period u (defined only for $t \in P_{iu}$)
$Y_{it} \in \{0, 1\}$	binary setup decision for item i in period t
$L_{iug} \in \{0, 1\}$	permission variable: $L_{iug} = 1$ authorises consumption from expiry group g when meeting demand (i, u)

Objective

$$\min\Big[\sum_{i \in I}\sum_{t \in T}\sum_{u \in P_{iu}} g_{itu}\,X_{itu} \;+\; \sum_{i \in I}\sum_{t \in T} s_{it}\,Y_{it}\Big]$$

Constraints

Global capacity For all $t \in T$,

$$\sum_{i \in I}\sum_{\substack{u \in P_{iu} \\ t \in P_{iu}}} X_{itu} \;\leq\; \kappa_t$$

Optional per-item capacity For all $i \in I$ and $t \in T$,

$$\sum_{u \in P_{iu}} X_{itu} \;\leq\; p_{it}$$

Setup linking For all $i \in I$ and $t \in T$,

$$\sum_{u \in P_{iu}} X_{itu} \;\leq\; \mu_{it}\,Y_{it}$$

Demand satisfaction For all $i \in I$ and $u \in T$,

$$\sum_{t \in P_{iu}} X_{itu} \;=\; d_{iu}$$

Last-expiry-first allocation For each $i \in I$ and $u \in T$ with $P_{iu} \neq \varnothing$, the following conditions apply. Let G_{iu} and $\mathcal{G}_{iug}$ be defined as above.

$$\sum_{t \in \mathcal{G}_{iug}} X_{itu} \;\leq\; C_{iu}\, L_{iug} \qquad \text{for each integer } g \text{ with } 0 \leq g < G_{iu} \tag{1a}$$

$$L_{iug} \;\geq\; L_{iug+1} \qquad \text{for each integer } g \text{ with } 0 \leq g < G_{iu} - 1 \tag{1b}$$

$$C_{iu}\left(1 - L_{iug+1}\right) \;\geq\; \sum_{h=0}^{g} \sum_{t \in \mathcal{G}_{iuh}} X_{itu} \;-\; d_{iu} \;+\; \varepsilon \quad \text{for each integer } g \text{ with } 0 \leq g < G_{iu} - 1 \tag{1c}$$

Variable domains

$$X_{itu} \geq 0, \quad Y_{it} \in \{0,1\}, \quad L_{iug} \in \{0,1\}$$

Notes

- The sequence $e_{iu}^{(0)} \geq e_{iu}^{(1)} \geq \cdots \geq e_{iu}^{(G_{iu}-1)}$ orders the expiration dates from latest to earliest, so group $g = 0$ contains the lots with the longest remaining life, while group $g = G_{iu} - 1$ contains those that expire next.
- The gate condition enforces that no flow is assigned from the group g unless the corresponding permission variable L_{iug} is set to 1.
- The monotonicity condition requires that an earlier expiring group cannot be used until all later expiring groups have been opened ($L_{iug} \geq L_{iug+1}$).
- The sufficiency cut closes group $g+1$ once the cumulative flow from groups $0, \ldots, g$ meets or exceeds the demand d_{iu}; otherwise the gate can open ($L_{iug+1} = 1$).